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EVEN END SEMESTER EXAMINATION – 2025

Name of the Course: B.Tech.- Aerospace Engineering

Semester: 4th

Paper Code: TAS-402

Name of the Paper: Aerodynamics I

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1		(20marks)
(a)	Explain the principle behind momentum equation. Derive the integral form of momentum equation.	CO2
(b)	i) What are the basic sources of Aerodynamics forces? Show using a neat figure these sources with their appropriate direction. ii) Define Vorticity and Kelvin's Theorem of Circulation	CO1
(c)	What do you mean by Euler equation? Derive Euler equation.	CO2
Q2		(20 marks)
(a)	i) Derive Laplace equation. ii) Explain elementary flows and their importance.	CO2
(b)	Theoretically prove that for a circular cylinder, $Cp=1-4\sin^2\Theta$.	CO2
(c)	Derive the equation of stream function and velocity potential function for a uniform flow with a velocity of V_∞. Show that the circulation in a uniform flow is zero.	CO3
Q3		(20 marks)
(a)	Derive an expression for the lift coefficient of a thin symmetric airfoil.	CO3
(b)	Explain vortex panel method to obtain lifting flow over arbitrary bodies. Note down two advantages and two disadvantages of vortex panel technique.	CO3
(c)	Write a short note on: i) Kutta Statement. ii) Starting Vortex.	CO3
Q4		(20 marks)
(a)	How the flow over a finite wing differ from flow over an airfoil? Explain with suitable diagram.	CO4
(b)	Using Biot-Savart Law derive an expression for the velocity induced by a semi-infinite vortex filament at a point in the flow.	CO4
(c)	Derive the fundamental equation of Prandtl's lifting line theory.	CO4
Q5		(20 marks)
(a)	Using the boundary layer theory, explain with suitable figures the flow physics over a golf ball and how it is different from a clean sphere?	CO5, CO6
(b)	Explain flow separation with neat sketches. Explain various methods used for preventing flow separation.	CO 5, CO6
(c)	Explain the different ways used for reduction of skin friction drag, Form drag and Induced drag	CO6